USB-2627 Multifunction DAQ Device - DATASHEET

Product Status: DISCONTINUED

Overview

The USB-2627 is a high-performance multifunction USB data acquisition device manufactured by Measurement Computing (MCC). It provides 64 single-ended or 32 differential analog input channels with 16-bit resolution and sample rates up to 1 MS/s. The device connects via Hi-Speed USB 2.0 and is bus-powered. Ideal for laboratory data collection, process monitoring, and industrial test applications. This product has been DISCONTINUED as of June 2025.

Analog Input Specifications

Resolution: 16 bits. Maximum sampling rate: 1 MS/s aggregate. Input ranges: +/-10V, +/-5V, +/-2V, +/-1V (software selectable per channel). Input impedance: 10 Gohm minimum. Absolute maximum input voltage: +/-20V. DNL: +/-1 LSB typical. INL: +/-2 LSB typical. Crosstalk: -85 dB at 100 kHz. Input FIFO: 128K samples. ADC type: SAR (successive approximation register).

Analog Output

Two 16-bit analog outputs with maximum update rate of 1 MS/s per channel. Output ranges: +/-10V. Settling time: 2 microseconds to +/-1 LSB. Output impedance: 0.2 ohm. Short-circuit current: +/-20 mA. Output FIFO depth: 8192 samples per channel for smooth waveform generation.

Digital I/O and Counters

48 digital I/O lines organized as six 8-bit ports (Port A through Port F). Each port individually configurable as input or output. Logic levels compatible with 3.3V and 5V systems. Four 32-bit counter/timer channels supporting event counting, period measurement, pulse width measurement, timing output, and encoder input (X1, X2, X4 quadrature). Maximum counter input frequency: 20 MHz.

Ordering and Replacement Notice

Part Number: USB-2627. Status: DISCONTINUED. Last order date: September 30, 2025. Recommended replacement: USB-2633 (next-generation model with USB 3.0, 2 MS/s sampling, and 4 analog outputs). Contact your MCC distributor for migration assistance.